

A Cross-Dataset Behavior-Aware Machine Learning Framework for Fuel Consumption Prediction and Fuel-Based CO₂ Emission Estimation

Abstract

Carbon dioxide emissions and fuel consumption are critical concerns in sustainable transportation, eco-driving systems, and intelligent transport systems. Most prior research on emission prediction relies on static vehicle certification data, traffic simulators, or fuel rating datasets, and few have combined driver behavior with fuel consumption estimates in a manner transferable across different datasets. This paper addresses that gap by introducing a behavior-aware machine learning framework that draws on both an OBD-II/CAN behavior classification dataset and the Vehicle Energy Dataset (VED) to predict fuel use and estimate CO₂ emissions.

We trained an OBD-II behavior classifier using vehicle speed, engine RPM, and absolute engine load as inputs. Evaluated via Leave-One-Driver-Out cross-validation, our Random Forest classifier achieved 97.62% accuracy and a macro F1-score of 96.56%. This trained classifier was then transferred to VED data, generating a behavior probability feature for each driving window. For fuel modeling, we focused on gasoline internal combustion engine vehicles and estimated fuel use using mass air flow (MAF) in 60-second windows; gasoline emission factors then yielded tailpipe CO₂ estimates.

Among all regression models evaluated, XGBoost performed best, achieving an R^2 of 0.8959, MAE of 0.9958 L/100 km, and RMSE of 1.3819 L/100 km. GroupKFold cross-validation confirmed robustness: $R^2 = 0.8852 \pm 0.0283$ and $\text{RMSE} = 1.4239 \pm 0.1949$ L/100 km. SHAP and ablation analyses identified speed, engine displacement, engine load, RPM, vehicle mass, and transferred behavior probability as the key predictors. Driving windows with higher risky-behavior scores consumed 15.19% more fuel and produced proportionally higher CO₂ emissions.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Transportation*; • **Information systems** → *Data analytics*.

Keywords

Fuel consumption prediction, CO₂ emission estimation, OBD-II, VED, driving behavior, XGBoost, SHAP, machine learning

1 Introduction

Fuel consumption and CO₂ emissions are central concerns in the push for sustainable transportation [1–3, 20]. Managing these challenges across eco-driving systems, intelligent transport networks, and real-world vehicle operation is far from straightforward. Emission levels depend not only on vehicle characteristics but also on dynamic factors such as speed, engine RPM, engine load, idling, acceleration, and overall driving style [4, 5]. This complexity makes machine learning a particularly valuable tool.

Prior work has explored fuel and CO₂ prediction across a wide range of data sources—certification records, OBD sensors, GPS traces, portable emission measurement systems (PEMS), vehicle-specific power metrics, and deep learning pipelines [1–8, 11, 15, 16]. While these studies demonstrate the viability of data-driven modeling, most depend on controlled test cycles that do not faithfully reflect real-world driving. Moreover, driving behavior and fuel consumption have largely been studied in isolation rather than jointly [9, 10, 17], leaving the predictive potential of OBD-II and CAN bus behavioral signals underutilized.

This study tackles that gap directly. We propose a cross-dataset, behavior-aware machine learning framework that draws on both an OBD-II/CAN behavior classification dataset and the Vehicle Energy Dataset (VED). The framework exploits shared features—vehicle speed, engine RPM, and engine load—across the two datasets. An OBD-II behavior classifier is first trained and validated using leave-one-driver-out cross-validation to ensure generalization across diverse driving styles [9, 10]. The classifier is then applied to VED data to label driving behaviors within each segment. Fuel consumption is derived from MAF measurements over 60-second windows, and CO₂ emissions are estimated from that fuel estimate [2, 11].

This approach bridges the gap between controlled laboratory data and real-world driving, and yields actionable insight into how driving style affects fuel use and emissions. The key contributions of this work are:

- A cross-dataset behavior-aware framework that links OBD-II/CAN behavioral data and VED to jointly predict fuel consumption and estimate fuel-based CO₂ emissions.
- Driving behavior classification using speed, engine RPM, and engine load, validated through leave-one-driver-out evaluation to account for diverse driving patterns.
- Transfer of behavior probabilities, Class-1 ratios, and OBD domain confidence scores into the VED fuel prediction pipeline, synchronized with 60-second windows.
- Rigorous evaluation via GroupKFold cross-validation with vehicle-specific splits, ablation studies, SHAP-based interpretability analysis, and statistical behavioral-score analysis.

2 Literature Review

Several machine learning models have been applied to forecasting CO₂ emissions and fuel consumption in vehicles. For instance, Udoh *et al.* developed decision tree, XGBoost, and linear regression models to predict CO₂ emissions in light-duty vehicles, though driving behavior was not considered [1]. Wang *et al.* incorporated driving behavior as a predictor in fuel consumption and CO₂ models, but the resulting models lacked interpretability [2]. Alam *et al.* and Yuan *et al.* proposed interpretable machine learning approaches for CO₂ emission prediction without accounting for driving behavior in fuel estimation [3, 4].

Table 1: Summary of Datasets and Their Roles

Dataset	Role	Key Variables	Output
OBD-II/CAN	Behavior learning	Speed, RPM, load, label, driver ID	Behavior classifier
VED dynamic	Fuel modeling	Timestamp, speed, MAF, RPM, load, trip ID	Window features
VED static	Metadata	Vehicle type, engine size, weight	Vehicle attributes

Real-world and sensor-based studies have used PEMS, GPS, OBD, vehicle-specific power, and trajectory data for fuel and emission estimation [5–8, 11, 18]. These studies demonstrate the value of dynamic vehicle data, but many are limited to specific scenarios—such as intersection modeling, GPS-speed estimation, or direct emission modeling—without cross-dataset behavioral transfer. Separately, driving behavior classification studies have shown that OBD-based speed, RPM, throttle, and engine-load features can effectively identify driving patterns [9, 10, 17]. However, classification outputs are rarely integrated into a unified fuel and CO₂ prediction pipeline.

In addition, hybrid and deep learning models have been developed for fuel consumption and emission estimation in transportation [12–14, 19], but these models focus solely on energy and emission prediction and do not explore behavioral transfer across independent datasets. Data-driven approaches using large-scale vehicle datasets have also demonstrated that integrating multiple signal types—speed profiles, engine parameters, and environmental context—improves both accuracy and generalizability [15, 16]. In summary, the literature reveals three persistent gaps: a lack of real-world vehicle behavior data in fuel models; the absence of a connection between behavioral classification and fuel/emission prediction; and the inability to perform cross-dataset behavioral transfer. This work addresses all three by training an OBD-II behavior classifier, transferring learned behaviors to VED, and rigorously evaluating results through ablation experiments, GroupKFold validation, SHAP analysis, and statistical behavior-score comparison.

3 Proposed Methodology

Fig. 1 illustrates the proposed cross-dataset behavior-aware framework, which consists of three major components: OBD-II driving behavior classification, VED fuel consumption modeling, and cross-dataset behavior-aware analysis.

3.1 Dataset Description

Two datasets were used in this study. The OBD-II/CAN behavioral driving classification dataset was used for behavioral analysis, while VED was used for fuel consumption prediction and CO₂ estimation. Because the two datasets were recorded on different vehicles, direct merging was not feasible; instead, common features were harmonized across both datasets.

The final VED subset was obtained by applying ICE gasoline filtering, MAF-based fuel target construction, 60-second time windowing, and OBD reliability filtering. Table 2 summarizes the resulting window-level dataset used for fuel and CO₂ estimation.

3.2 OBD-II Driving Behavior Classification

OBD-II dataset preprocessing involved column renaming, numeric conversion, feature selection, and physical feasibility filtering. The common features selected were vehicle speed, engine RPM, and

Table 2: Final VED Reliable Window Dataset Summary

Item	Value
Reliable windows	66,927
Vehicles	158
Trips	11,730
Window size	60 seconds
Target	MAF-derived fuel consumption
Fuel unit	L/100 km
CO ₂ unit	g/km
Reliability threshold	OBD-domain ratio ≥ 0.80

absolute engine load. Since the OBD-II dataset contained no time or trip information, classification was performed row-wise.

Machine learning models evaluated included Logistic Regression, Decision Tree, Random Forest, Hist Gradient Boosting, and XGBoost. Leave-One-Driver-Out (LODO) cross-validation was used to mitigate per-driver bias: at each fold, one driver’s data formed the test set while the remaining drivers’ data formed the training set.

3.3 VED Preprocessing and Vehicle Filtering

VED dynamic data files were processed individually to reduce memory usage, and static vehicle metadata were joined using vehicle IDs. Only gasoline internal combustion engine vehicles were retained; diesel, flex-fuel, electric, and plug-in hybrid vehicles were excluded to ensure consistency in fuel and CO₂ calculations.

Dynamic data were sorted by vehicle ID, trip ID, and timestamp, with duplicate timestamps removed. The time interval was computed as

$$\Delta t = \frac{t_i - t_{i-1}}{1000}. \quad (1)$$

Distance per sample was computed as

$$d_i = \frac{v_i \Delta t_i}{3600}, \quad (2)$$

where v_i is vehicle speed in km/h.

3.4 MAF-Derived Fuel Target Construction

Reported fuel-rate observations were not usable in the selected gasoline ICE subset alongside the behavior-transfer features. MAF-derived fuel consumption was therefore used as the prediction target. Fuel mass flow was computed as

$$\dot{m}_{\text{fuel}} = \frac{\text{MAF}}{14.7}, \quad (3)$$

where 14.7 is the stoichiometric gasoline air–fuel ratio. Fuel volume flow was then calculated as

$$\dot{V}_{\text{fuel}} = \frac{\dot{m}_{\text{fuel}}}{745}, \quad (4)$$

where 745 g/L is gasoline density. Fuel consumed per sample was

$$\text{Fuel}_i = \dot{V}_{\text{fuel},i} \times \Delta t_i. \quad (5)$$

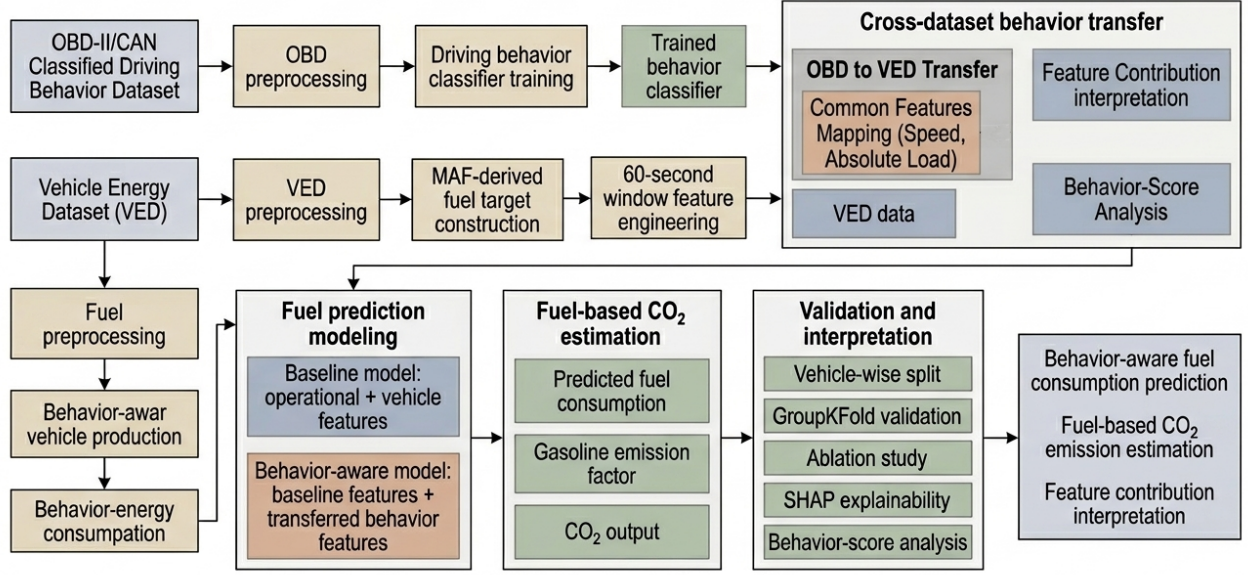


Figure 1: Proposed cross-dataset behavior-aware methodology for fuel consumption prediction and fuel-based CO₂ emission estimation.

The final window-level fuel consumption target was

$$FC = \frac{\sum \text{Fuel}_i}{\sum d_i} \times 100. \quad (6)$$

MAF was excluded from model input features to avoid target leakage.

3.5 Cross-Dataset Behavior Transfer

The Random Forest classifier trained on the OBD-II dataset—using speed, RPM, and absolute engine load—was applied to VED windows. Its output, namely the Class-1 probability and Class-1 ratio, was used as transferred behavior features. Since no ground-truth behavior labels exist in VED, these outputs serve as behavior predictions. Windows were further filtered using an OBD-domain ratio threshold of 0.80.

3.6 Window-Level Feature Engineering

VED data were aggregated into 60-second windows. Each window comprised statistics on speed, RPM, load, acceleration, idling ratio, harsh acceleration count, harsh deceleration count, vehicle mass, engine displacement, transferred behavior probability, Class-1 ratio, and OBD domain ratio. A total of 66,927 windows were extracted from 11,730 trips by 158 vehicles.

3.7 Fuel-Based CO₂ Estimation

CO₂ emissions were estimated from predicted fuel consumption using

$$CO_2 = FC \times 23.49, \quad (7)$$

where FC is fuel consumption in L/100 km. The resulting CO₂ values represent fuel-based tailpipe estimates rather than directly measured exhaust emissions.

Table 3: XGBoost Hyperparameter Tuning Search Space

Hyperparameter	Candidate Values
n_estimators	300, 500, 800
max_depth	3, 4, 5, 6
learning_rate	0.01, 0.03, 0.05, 0.10
subsample	0.7, 0.8, 0.9, 1.0
colsample_bytree	0.7, 0.8, 0.9, 1.0
reg_alpha	0, 0.01, 0.1
reg_lambda	1, 5, 10

4 Experimental Setup

Two fuel prediction configurations were evaluated. The baseline configuration used only vehicle operational features and static meta-data. The behavior-aware configuration additionally incorporated transferred behavior probability, Class-1 ratio, and OBD domain reliability. Regression algorithms tested were: Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, Extra Trees, Hist Gradient Boosting, and XGBoost.

Hyperparameter tuning for the best-performing behavior-aware XGBoost model was conducted via RandomizedSearchCV with GroupKFold cross-validation, grouping by vehicle ID. Table 3 lists the search space.

Vehicle-wise train-test splitting was used to prevent data leakage across vehicles. GroupKFold cross-validation was applied with

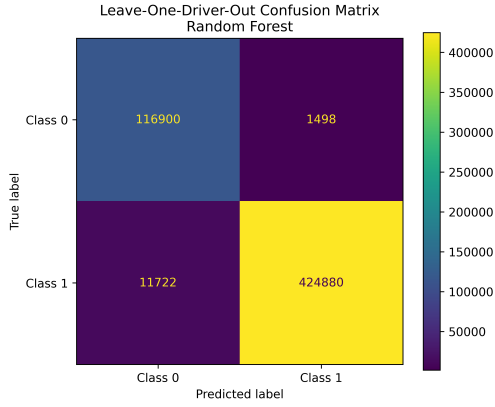


Figure 2: Leave-One-Driver-Out confusion matrix of the Random Forest driving behavior classifier.

Table 4: OBD-II Driving Behavior Classification Result

Model	Acc.	Prec.	Recall	F1
Random Forest	0.9762	0.9527	0.9802	0.9656

vehicle ID as the grouping variable. Evaluation metrics were R^2 , mean absolute error (MAE), and root mean squared error (RMSE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (8)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (9)$$

5 Results and Discussion

5.1 Driving Behavior Classification Results

Random Forest achieved the highest OBD-II classification performance, with 97.62% accuracy and a macro F1-score of 96.56% under Leave-One-Driver-Out cross-validation, demonstrating that the selected feature set learns transferable driving behavior patterns.

Fig. 2 shows that the classifier correctly identified most samples from both behavior classes, supporting its use for transferring behavior probability features to VED.

5.2 Fuel Prediction Results

Table 5 presents the best baseline and behavior-aware fuel prediction results. The behavior-aware XGBoost model achieved the best overall performance with $R^2 = 0.8959$, MAE = 0.9958 L/100 km, and RMSE = 1.3819 L/100 km. The corresponding CO₂ estimation yielded MAE = 23.39 g/km and RMSE = 32.46 g/km.

5.3 GroupKFold Cross-Validation

Vehicle-wise GroupKFold validation confirmed model robustness. The behavior-aware XGBoost model achieved mean $R^2 = 0.8852 \pm 0.0283$ and RMSE = 1.4239 ± 0.1949 L/100 km.

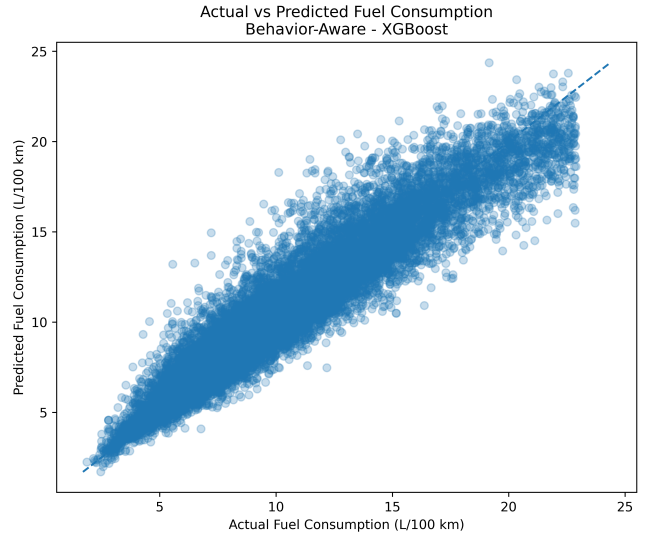


Figure 3: Actual versus predicted fuel consumption for the behavior-aware XGBoost model.

Table 5: Best Baseline and Behavior-Aware Fuel Prediction Results

Setup	Model	R^2	MAE	RMSE
Baseline	HGB	0.8951	1.0036	1.3871
Behavior-aware	XGBoost	0.8959	0.9958	1.3819

Table 6: Vehicle-Wise GroupKFold Cross-Validation Results

Setup	R^2	MAE	RMSE
Baseline XGBoost	0.8846 ± 0.0288	1.0270 ± 0.1409	1.4273 ± 0.1986
Behavior-aware XGBoost	0.8852 ± 0.0283	1.0216 ± 0.1387	1.4239 ± 0.1949

Table 7: Ablation Study Using XGBoost

Feature Set	No. Features	R^2	MAE	RMSE
A1: Operation only	19	0.7094	1.7566	2.3087
A2: + Metadata	21	0.8949	1.0077	1.3884
A3: + Behavior	23	0.8952	0.9995	1.3867
A4: + Reliability	24	0.8959	0.9958	1.3819

5.4 Ablation Study

Table 7 reports the results of ablation experiments conducted on XGBoost. Operational features alone yielded notably weaker performance. Adding vehicle metadata produced the largest single improvement in RMSE, while the incremental addition of behavioral and reliability features further refined predictions.

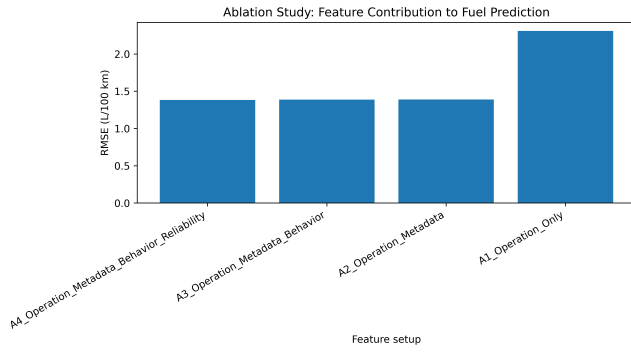


Figure 4: Ablation study showing the contribution of operational, vehicle metadata, transferred behavior, and reliability features.

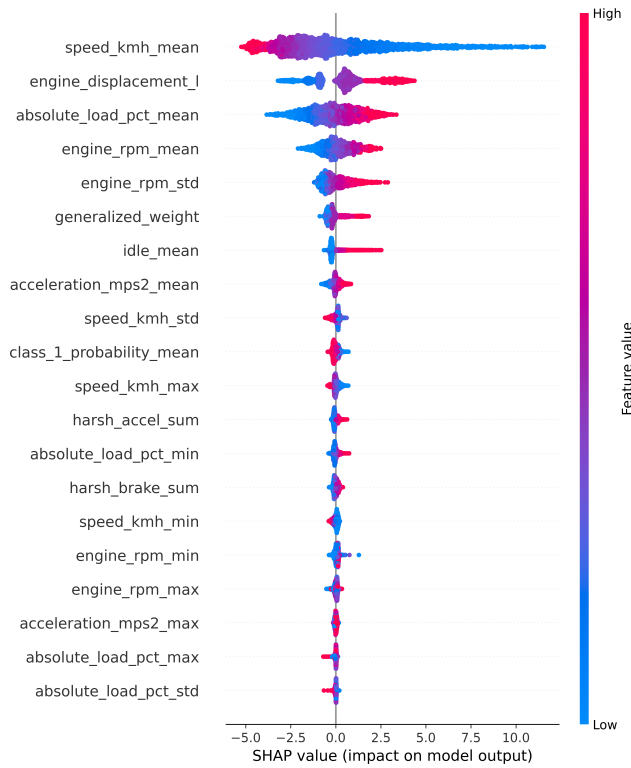


Figure 5: SHAP summary plot for the behavior-aware XGBoost fuel prediction model.

5.5 SHAP Explainability Analysis

SHAP analysis was performed on the behavior-aware XGBoost model. The ten most influential features were average speed, engine displacement, mean engine load, mean RPM, RPM variability, vehicle mass, idle ratio, mean acceleration, speed variability, and transferred behavior probability. The presence of transferred behavior probability among the top ten confirms that behavioral information adds predictive value.

Table 8: Top SHAP Feature Importance Values

Feature	Mean SHAP
speed_kmh_mean	2.5202
engine_displacement_l	1.3547
absolute_load_pct_mean	1.0978
engine_rpm_mean	0.6589
engine_rpm_std	0.5828
generalized_weight	0.3708
idle_mean	0.3147
acceleration_mps2_mean	0.1903
speed_kmh_std	0.1453
class_1_probability_mean	0.1156

Table 9: Low vs. High Transferred Behavior-Score Comparison

Group	N	Fuel (L/100 km)	CO ₂ (g/km)
Low score	16,732	9.5658	224.70
High score	16,732	11.0189	258.83

5.6 Behavior-Score Statistical Analysis

A comparison between the lowest and highest quartiles of transferred Class-1 probability was conducted. The mean fuel consumption for the high behavior-score group was 11.0189 L/100 km versus 9.5658 L/100 km for the low-score group, representing a 15.19% difference in fuel consumption and fuel-based CO₂ emissions.

A Mann-Whitney U test confirmed that the difference in fuel consumption and CO₂ emissions between the two groups was statistically significant ($p = 2.48 \times 10^{-127}$). However, Cliff's delta and Spearman's correlation were 0.1428 and 0.1013, respectively, indicating a small practical effect size.

5.7 Discussion of Key Findings

The results confirm that core operational parameters (speed, RPM, load) and vehicle metadata are the primary drivers of fuel prediction from MAF-derived targets, as evidenced by the large RMSE reduction when metadata are added to operational features in the ablation study. Behavioral features provide an additional, statistically significant contribution, a finding supported by both the ablation results and the SHAP analysis, which places transferred behavior probability among the top ten predictors, albeit below average speed, engine displacement, engine load, and RPM in importance.

The behavior-score analysis further validates the behavior-aware model. Windows with higher transferred Class-1 scores consistently exhibited greater fuel consumption and CO₂ emissions. The small

Spearman coefficient and Cliff's delta indicate, however, that transferred behavioral data should be treated as a supportive rather than dominant signal. The framework is therefore accurately characterized as behavior-aware rather than behavior-driven.

6 Limitations

Several limitations should be noted. First, because actual fuel-rate values could not be used in the selected gasoline ICE subset, the fuel target was derived from MAF measurements; the reported fuel consumption figures therefore reflect MAF-based estimates rather than directly recorded fuel use. Second, CO₂ emissions were not measured with PEMS but were estimated from fuel consumption using a fixed emission factor. Third, vehicle behavior labels in VED are classifier-predicted rather than ground-truth. Finally, the marginal effect of behavioral features suggests that core vehicle operational data remains the primary determinant of predicted emissions.

7 Conclusion and Future Work

This study introduced a cross-dataset, behavior-aware machine learning framework for predicting fuel consumption and estimating fuel-based CO₂ emissions. An OBD-II-trained behavior classifier, using speed, RPM, and absolute engine load, was transferred to VED data to generate behavioral probability features. The behavior-aware XGBoost model achieved the best fuel prediction performance ($R^2 = 0.8959$, MAE = 0.9958 L/100 km, RMSE = 1.3819 L/100 km), with GroupKFold validation confirming robustness ($R^2 = 0.8852$, RMSE = 1.4239 L/100 km on average). SHAP and ablation analyses identified speed, engine displacement, engine load, RPM, vehicle mass, and transferred behavior probability as the most influential predictors. High behavior-score windows exhibited 15.19% greater fuel consumption and CO₂ emissions compared to low behavior-score windows.

Future work will validate the framework using directly measured fuel-rate and PEMS-based CO₂ data. Additional contextual variables—road grade, traffic conditions, road type, and weather—may further improve generalization. Sequence-based deep learning architectures, including LSTM, GRU, 1D-CNN, and transformer models, will also be explored to capture temporal driving dynamics from raw time-series data.

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